

Terraform-01

1). Install Terraform on your PC

Download terraform:

Extract file:

▼ Today (4)

terraform_1.15.5_windows_amd64	6/9/2026 3:18 PM	File folder	
ECS - TASKS	6/9/2026 7:13 AM	Office Open XML ...	4,636 KB
ECS Task	6/9/2026 8:06 AM	Microsoft Edge PD...	3,305 KB
terraform_1.15.5_windows_amd64	6/9/2026 3:16 PM	WinRAR ZIP archive	34,815 KB

▼ Earlier this week (16)

Copy to C Drive

← → ▼ ↑ > This PC > Local Disk (C:) > terraform

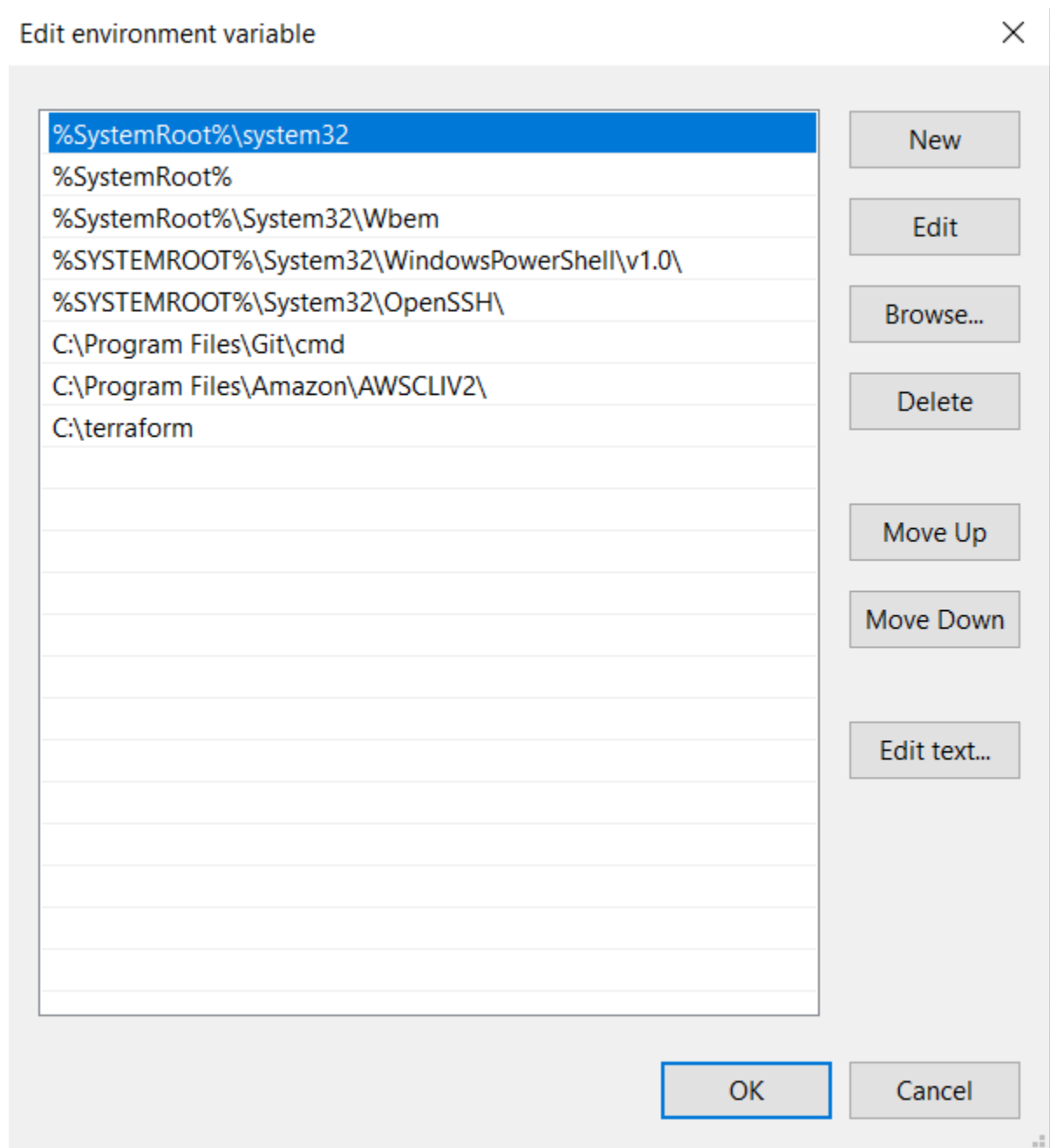
	Name	Date modified	Type	Size
▼ Quick access				
Desktop	LICENSE	5/27/2026 6:20 PM	Text Document	5 KB
Downloads	terraform	5/27/2026 6:20 PM	Application	116,934 KB
Documents				
Pictures				
This PC				

Check the version:

```
DELL@DESKTOP-B7FOCBT MINGW64 /c/terraform
$ terraform -v
Terraform v1.15.5
on windows_amd64

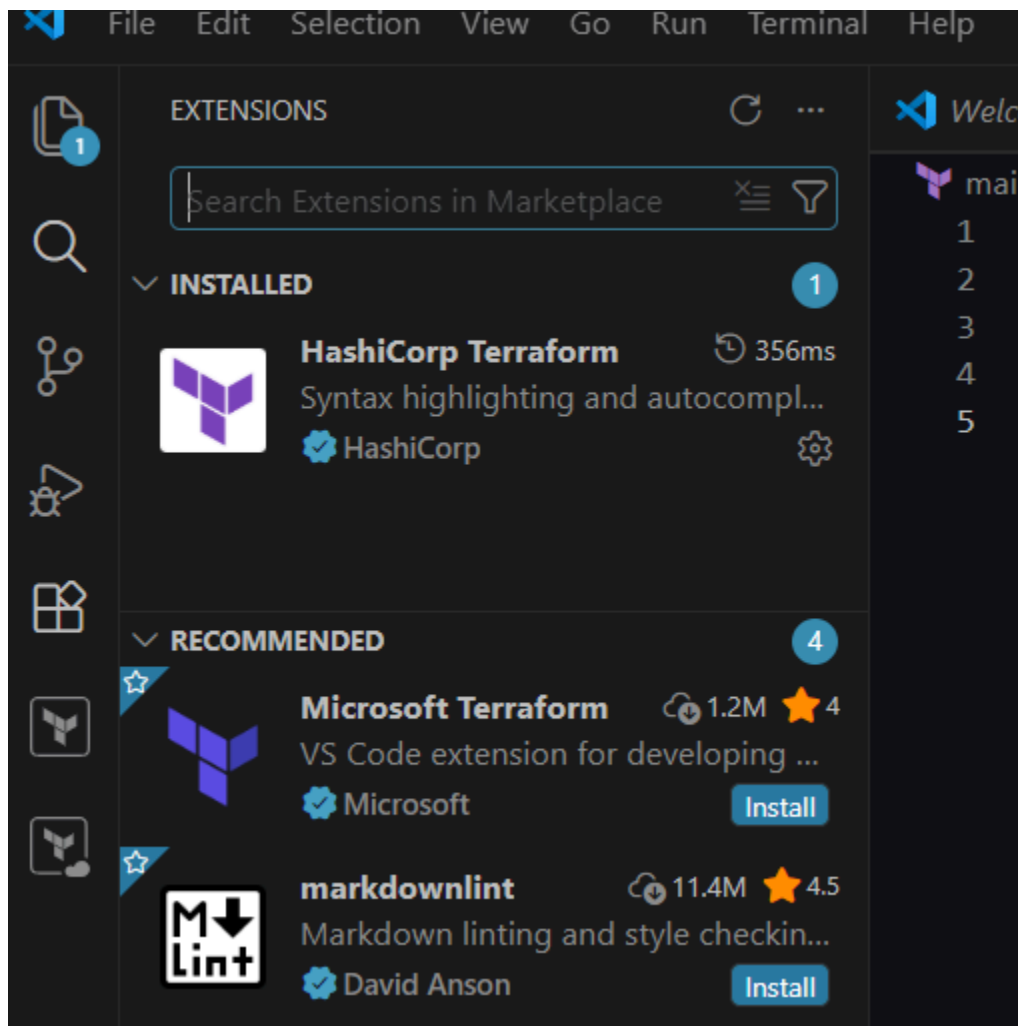
DELL@DESKTOP-B7FOCBT MINGW64 /c/terraform
$ |
```

Set env variables:



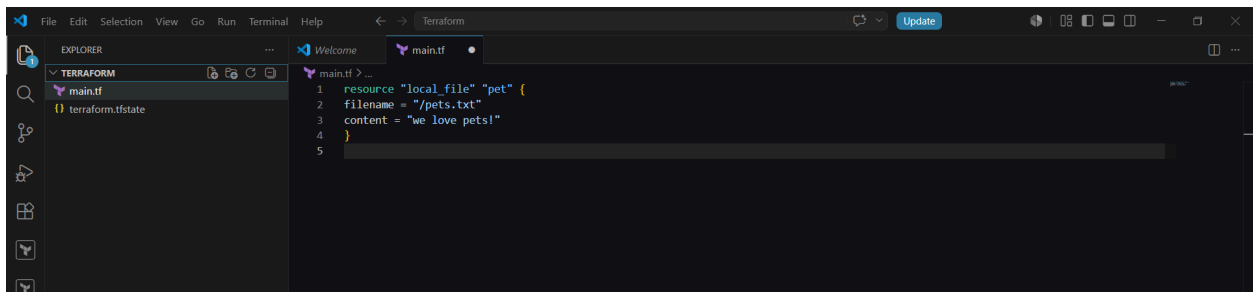
2). Execute all the templates shown in the video.

Open visual studio code and download terraform extensions

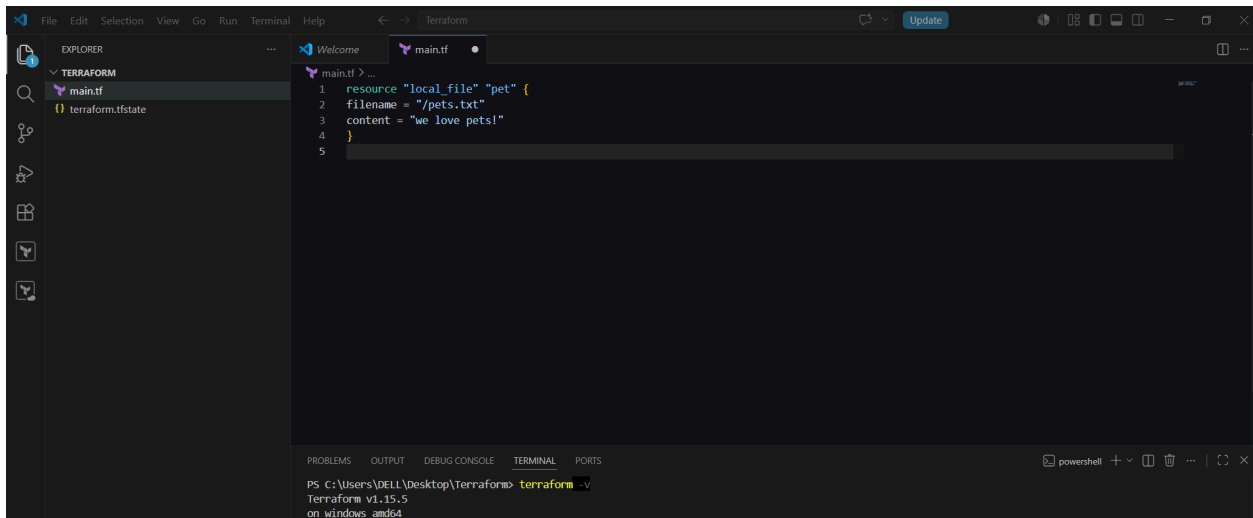


Create terraform folder
Inside folder create [main.tf](#) file

```
resource "local_file" "pet" {  
  filename = "/pets.txt"  
  content = "we love pets!"  
}
```



Open terminal:
Run the commands
Check the version: terraform -v



terraform init: Initializes Terraform working directory

```
PS C:\Users\DELL\Desktop\Terraform> terraform init
Initializing provider plugins found in the configuration...

Initializing the backend...

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

terraform plan: Shows **execution plan before applying changes**

```
PS C:\Users\DELL\Desktop\Terraform> terraform plan

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes
are needed.
```

Terraform apply: Executes the changes defined in configuration

```
PS C:\Users\DELL\Desktop\Terraform> terraform apply

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes
are needed.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.
PS C:\Users\DELL\Desktop\Terraform> █
```

3). Note down below points, Terraform Init Terraform Plan
Terraform Apply Terraform Provider.

1.Terraform Init

terraform init

Purpose:

- Initializes Terraform working directory
- Downloads required **provider plugins**
- Sets up backend configuration (if any)

What happens internally:

- Creates `.terraform/` folder
 - Downloads provider (AWS, local, etc.)
 - Reads configuration files
-

2.Terraform Plan

`terraform plan`

Purpose:

- Shows **execution plan before applying changes**

What it does:

- Compares current state vs desired state
- Shows:
 - What will be created (+)
 - What will be changed (~)
 - What will be destroyed (-)

Use:

Safe preview step before making real changes

3.Terraform Apply

`terraform apply`

or

`terraform apply -auto-approve`

Purpose:

- Executes the changes defined in configuration

What it does:

- Creates real infrastructure (EC2, files, VPC, etc.)
 - Updates Terraform state file (`terraform.tfstate`)
-

4. Terraform Provider

Definition:

A provider is a plugin that allows Terraform to interact with APIs

Example:

```
provider "aws" {  
  region = "ap-south-1"  
}
```

What it does:

- Connects Terraform to cloud platforms
- Examples:
 - AWS Provider → AWS services
 - Azure Provider → Azure resources
 - Local Provider → local files

4). Integrate a sample Terraform template in Jenkins.

```
[root@ip-172-31-14-229 ~]# systemctl start jenkins
[root@ip-172-31-14-229 ~]# systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; disabled; preset: disabled)
   Active: active (running) since Tue 2026-06-09 10:15:12 UTC; 4s ago
     Main PID: 3042 (java)
       Tasks: 47 (limit: 4543)
      Memory: 620.7M
         CPU: 20.731s
    CGroup: /system.slice/jenkins.service
            └─3042 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

Jun 09 10:15:10 ip-172-31-14-229.ap-south-1.compute.internal jenkins[3042]: 2026-06-09 10:15:10.518+0000 [id=36] INFO jenkins.InitReactorRunner$1#onAttain
Jun 09 10:15:11 ip-172-31-14-229.ap-south-1.compute.internal jenkins[3042]: 2026-06-09 10:15:11.755+0000 [id=37] INFO h.p.b.g.GlobalTimeoutConfiguration#l
Jun 09 10:15:12 ip-172-31-14-229.ap-south-1.compute.internal jenkins[3042]: 2026-06-09 10:15:12.535+0000 [id=37] INFO jenkins.InitReactorRunner$1#onAttain
Jun 09 10:15:12 ip-172-31-14-229.ap-south-1.compute.internal jenkins[3042]: 2026-06-09 10:15:12.535+0000 [id=36] INFO jenkins.InitReactorRunner$1#onAttain
Jun 09 10:15:12 ip-172-31-14-229.ap-south-1.compute.internal jenkins[3042]: 2026-06-09 10:15:12.625+0000 [id=38] INFO jenkins.InitReactorRunner$1#onAttain
Jun 09 10:15:12 ip-172-31-14-229.ap-south-1.compute.internal jenkins[3042]: 2026-06-09 10:15:12.648+0000 [id=36] INFO jenkins.InitReactorRunner$1#onAttain
Jun 09 10:15:12 ip-172-31-14-229.ap-south-1.compute.internal jenkins[3042]: 2026-06-09 10:15:12.668+0000 [id=56] INFO hudson.util.Retrier#start: Attempt #
Jun 09 10:15:12 ip-172-31-14-229.ap-south-1.compute.internal jenkins[3042]: 2026-06-09 10:15:12.738+0000 [id=37] INFO jenkins.InitReactorRunner$1#onAttain
Jun 09 10:15:12 ip-172-31-14-229.ap-south-1.compute.internal jenkins[3042]: 2026-06-09 10:15:12.781+0000 [id=30] INFO hudson.lifecycle.Lifecycle#onReady:
Jun 09 10:15:12 ip-172-31-14-229.ap-south-1.compute.internal systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
```

Install terraform: `sudo yum install -y yum-utils`
`sudo yum-config-manager --add-repo`
<https://rpm.releases.hashicorp.com/AmazonLinux/hashicorp.repo>
`sudo yum install terraform -y`

```
[root@ip-172-31-14-229 ~]# sudo yum install -y yum-utils
sudo yum-config-manager --add-repo https://rpm.releases.hashicorp.com/AmazonLinux/hashicorp.repo
sudo yum install terraform -y
Hashicorp Stable - x86_64
Package dnf-utils-4.3.0-13.amzn2023.0.6.noarch is already installed.
Dependencies resolved.
Nothing to do.
Complete!
Adding repo from: https://rpm.releases.hashicorp.com/AmazonLinux/hashicorp.repo
Last metadata expiration check: 0:00:01 ago on Tue Jun 9 10:29:43 2026.
Dependencies resolved.
```

Package	Architecture	Version	Repository
Installing: terraform	x86_64	1.15.5-1	hashicorp

```
Transaction Summary
-----
Install 1 Package
```

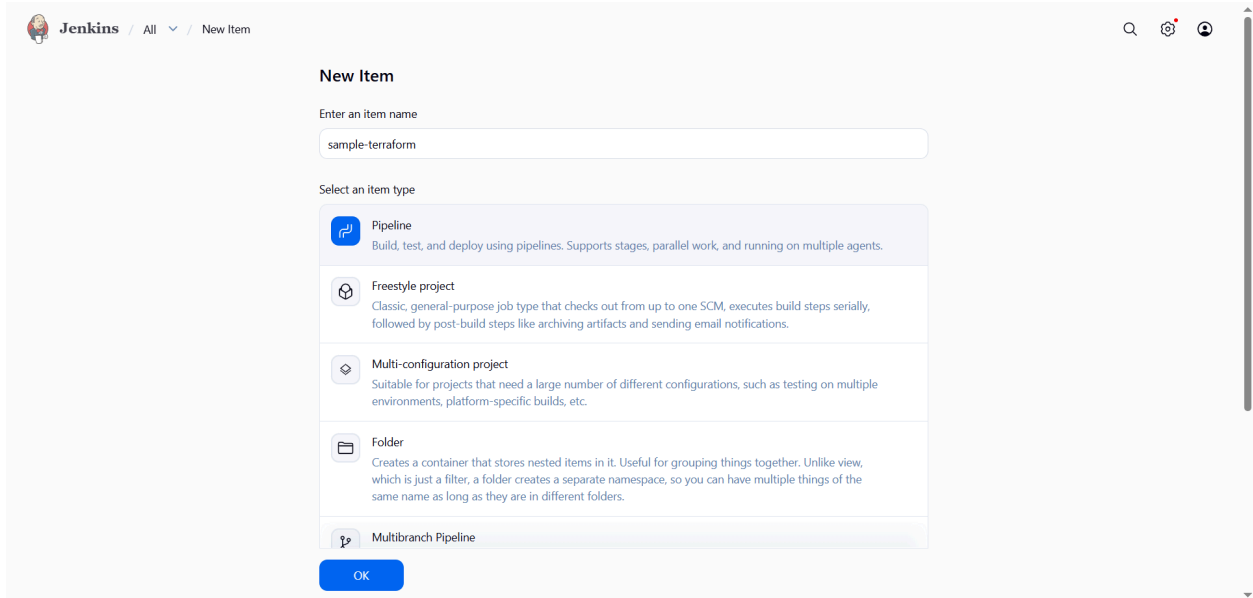
Check version:

```
Complete!
[root@ip-172-31-14-229 ~]# terraform --version
Terraform v1.15.5
on linux_amd64
[root@ip-172-31-14-229 ~]# █
```

i-0532703582a1480a1 (Jenkins)

PublicIPs: 43.204.111.74 PrivateIPs: 172.31.14.229

Create new item :



Create main.tf

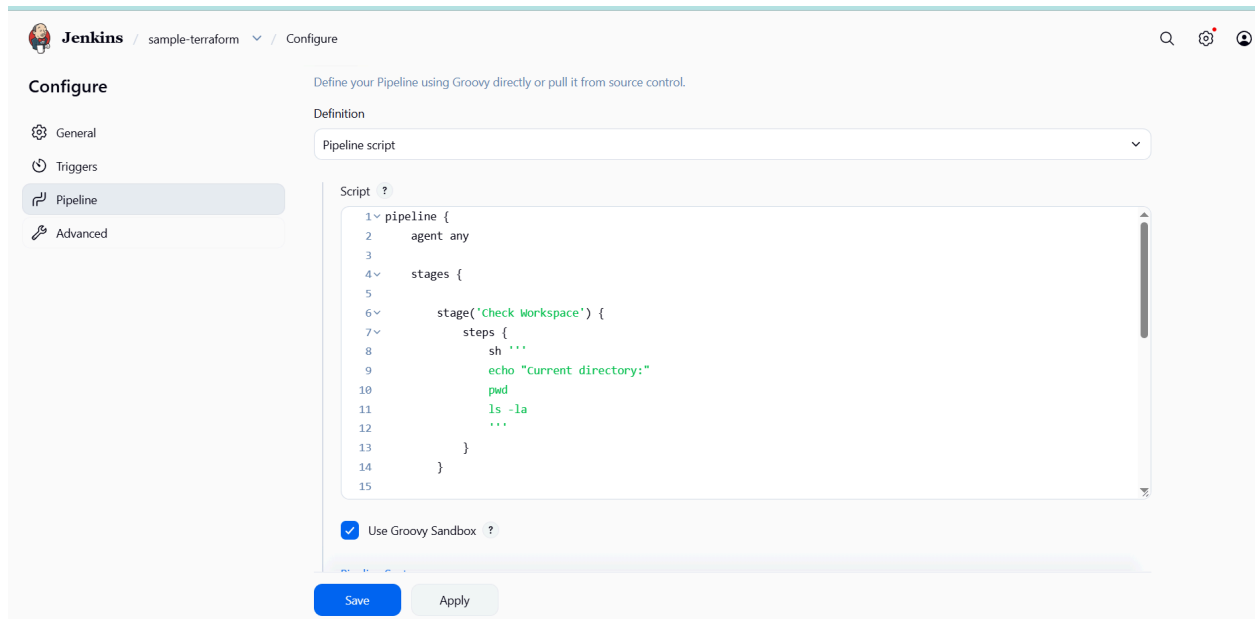
```
[root@ip-172-31-14-229 sample-terraform]# pwd
/var/lib/jenkins/workspace/sample-terraform
[root@ip-172-31-14-229 sample-terraform]# cat main.tf
resource "local_file" "pet" {
  filename = "pets.txt"
  content = "we love pets!"
}

[root@ip-172-31-14-229 sample-terraform]#
```

i-0532703582a1480a1 (Jenkins)

PublicIPs: 43.204.111.74 PrivateIPs: 172.31.14.229

Write a pipeline:



```
pipeline {  
  agent any
```

```
  stages {
```

```
    stage('Check Workspace') {  
      steps {  
        sh ""  
        echo "Current directory:"  
        pwd  
        ls -la  
        ""  
      }  
    }  
  }
```

```
    stage('Terraform Init') {  
      steps {  
        sh 'terraform init'  
      }  
    }
```

```

    }

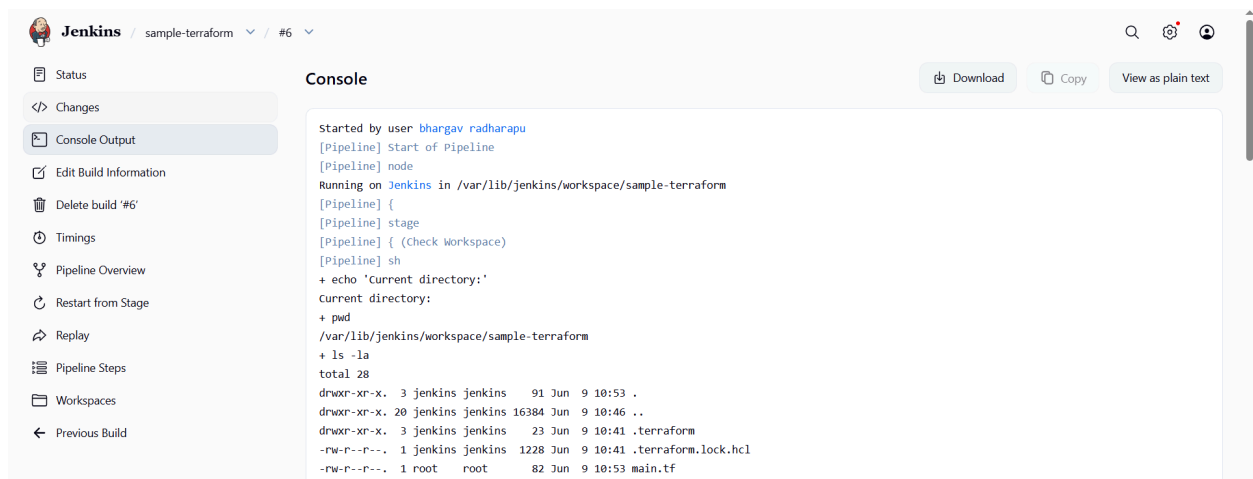
    stage('Terraform Plan') {
        steps {
            sh 'terraform plan'
        }
    }

    stage('Terraform Apply') {
        steps {
            sh 'terraform apply -auto-approve'
        }
    }
}

```

Save & build

Console output:



The screenshot shows the Jenkins web interface for a pipeline named 'sample-terraform' at build #6. The left sidebar contains navigation options: Status, Changes, Console Output (selected), Edit Build Information, Delete build '#6', Timings, Pipeline Overview, Restart from Stage, Replay, Pipeline Steps, Workspaces, and Previous Build. The main console area displays the following output:

```

Started by user bhargav radharapu
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/lib/jenkins/workspace/sample-terraform
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Check Workspace)
[Pipeline] sh
+ echo 'Current directory:'
Current directory:
+ pwd
/var/lib/jenkins/workspace/sample-terraform
+ ls -la
total 28
drwxr-xr-x. 3 jenkins jenkins  91 Jun  9 10:53 .
drwxr-xr-x. 20 jenkins jenkins 16384 Jun  9 10:46 ..
drwxr-xr-x. 3 jenkins jenkins  23 Jun  9 10:41 .terraform
-rw-r--r--. 1 jenkins jenkins 1228 Jun  9 10:41 .terraform.lock.hcl
-rw-r--r--. 1 root   root      82 Jun  9 10:53 main.tf

```

At the top right of the console area, there are buttons for 'Download', 'Copy', and 'View as plain text'.



```
    [32m+[0m[0m content_base64sha256 = (known after apply)
    [32m+[0m[0m content_base64sha512 = (known after apply)
    [32m+[0m[0m content_md5           = (known after apply)
    [32m+[0m[0m content_sha1          = (known after apply)
    [32m+[0m[0m content_sha256        = (known after apply)
    [32m+[0m[0m content_sha512        = (known after apply)
    [32m+[0m[0m directory_permission = "0777"
    [32m+[0m[0m file_permission       = "0777"
    [32m+[0m[0m filename              = "pets.txt"
    [32m+[0m[0m id                   = (known after apply)
  }

[1mPlan:[0m [0m1 to add, 0 to change, 0 to destroy.
[0m[1mlocal_file.pet: Creating...[0m[0m
[0m[1mlocal_file.pet: Creation complete after 0s [id=fefacccdae259f25533749abfb90e27558256459][0m
[0m[1m[32m
Apply complete! Resources: 1 added, 0 changed, 0 destroyed.[0m
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```